STACKS, MODULI OF CURVES: LEMMA-CONTRACT-PRESTABLE-TO-STABLE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3126-3140 of sources/stacks-moduli-curves/pinned-original-source.tex; proof span is lines 3142-3192, the complete author proof (49 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 6 of queue_moduli-curves.json (pb=2307, stmt 3126-3140, proof 3142-3192).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0E7Q: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3192–3192. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E7N: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3192–3192. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E88: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2865–2874. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E7C: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2987–2995. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E89: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2900–2912. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E76: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2679–2688. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E78: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2736–2749. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a prestable family of curves of genus $g \geq 2$. There is a factorization $X \to Y \to S$ of f where $g : Y \to S$ is a stable family of curves and $c : X \to Y$ has the following properties*

(1) *$\mathcal{O}_Y = c_*\mathcal{O}_X$ and $R^1c_*\mathcal{O}_X = 0$ and this remains true after base change by any morphism $S' \to S$, and*
(2) *for any $s \in S$ the morphism $c_s : X_s \to Y_s$ is the contraction of rational tails and bridges discussed in Algebraic Curves, Section 0E7N.*

Moreover $c : X \to Y$ is unique up to unique isomorphism.

Proof. Let $s \in S$. Let $c_0 : X_s \to Y_0$ be the contraction of Algebraic Curves, Section 0E7N (more precisely Algebraic Curves, Lemma 0E7Q). By Lemma 0E7C there exists an elementary étale neighbourhood (U, u) and a morphism $c : X_U \to Y$ of families of curves over U which recovers c_0 as the fibre at u. Since ω_{Y_0} is ample, after possibly shrinking U, we see that $Y \to U$ is a stable family of genus g by the openness inherent in Lemmas 0E76 and 0E78. After possibly shrinking U once more, assertion (1) of the lemma for $c : X_U \to Y$ follows from Lemma 0E88. Moreover, part (2) holds by the uniqueness in Algebraic Curves, Lemma 0E7Q. We conclude that a morphism c as in the lemma exists étale locally on S. More precisely, there exists an étale covering $\{U_i \to S\}$ and morphisms $c_i : X_{U_i} \to Y_i$ over U_i where $Y_i \to U_i$ is a stable family of curves having properties (1) and (2) stated in the lemma.

To finish the proof it suffices to prove uniqueness of $c : X \to Y$ (up to unique isomorphism). Namely, once this is done, then we obtain isomorphisms

$$\varphi_{ij} : Y_i \times_{U_i} (U_i \times_S U_j) \longrightarrow Y_i \times_{U_j} (U_i \times_S U_j)$$

satisfying the cocycle condition (by uniqueness) over $U_i \times U_j \times U_k$. Since $\overline{\mathcal{M}_g}$ is an algebraic stack, we have effectiveness of descent data and we obtain $Y \to S$. The morphisms c_i descend to a morphism $c : X \to Y$ over S. Finally, properties (1) and (2) for c are immediate from properties (1) and (2) for c_i.

Finally, if $c_1 : X \to Y_i$, $i = 1, 2$ are two morphisms towards stably families of curves over S satisfying (1) and (2), then we obtain a morphism $Y_1 \to Y_2$ compatible with c_1 and c_2 at least locally on S by Lemma 0E89. We omit the verification that these morphisms are unique (hint: this follows from the fact that the scheme theoretic image of c_1 is Y_1). Hence these locally given morphisms glue and the proof is complete. □